3. Fase Práctica: *Hardening* y Vigilancia en la Trinchera

1. **Generación de ataques controlados**.

```
pablo@pablo:/etc/ansible$ ssh administrador_falso@192.168.20.165
The authenticity of host '192.168.20.165 (192.168.20.165)' can't be established.
ED25519 key fingerprint is SHA256:twleUBuJhh3bhDG7cgPQKZesUOUpCsT3Qv1JIgFNMzw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.20.165' (ED25519) to the list of known hosts.
administrador_falso@192.168.20.165's password:
Permission denied, please try again.
administrador_falso@192.168.20.165's password:
Permission denied, please try again.
administrador_falso@192.168.20.165's password:
administrador_falso@192.168.20.165: Permission denied (publickey,password).
```

2. **Análisis Forense Manual**.

```
pablo@pablo:/etc/ansible$ sudo tail -n 50 /var/log/auth.log | grep "administrador_falso"
2026-04-17T12:22:51.488064+00:00 pablo sshd[2686]: Invalid user administrador_falso from 192.168.20.165 port 56850
2026-04-17T12:23:13.122566+00:00 pablo sshd[2686]: Failed password for invalid user administrador_falso from 192.168.20.165 port 56850 ssh2
2026-04-17T12:23:17.251312+00:00 pablo sshd[2686]: Failed password for invalid user administrador_falso from 192.168.20.165 port 56850 ssh2
2026-04-17T12:23:22.022352+00:00 pablo sshd[2686]: Failed password for invalid user administrador_falso from 192.168.20.165 port 56850 ssh2
2026-04-17T12:23:23.538296+00:00 pablo sshd[2686]: Connection closed by invalid user administrador_falso 192.168.20.165 port 56850 [preauth]
pablo@pablo:/etc/ansible$ _
```

3. **Automatización inteligente con Bash**

```
  GNU nano 7.2                                                    check_intruders.sh *
#!/bin/bash
LOG_FILE="/var/log/auth.log"
REPORT_FILE="alertas.txt"
CURRENT_DATE=$(date+"%Y-%,-%d  %H:%M:%S")
TODAY_SYSLOG=$(date +"%b %e")
echo "=====================================================" > "$REPORT_FILE"
echo "REPORTE DE INTRUSIONES -Generado el. $CURRENT_DATE" >> "$REPORT_FILE"
echo "=====================================================" >"$REPORT_FILE"
echo "">> "$REPORT_FILE"

TOTAL_ERRORS=$(grep "$TODAY_SYSLOG" "$LOG_FILE" | grep "Failed password" |wc -l)

echo "Se han detectadp un total de $TOTAL_ERRORS intentos fallidos de conexion hoy."> "$REPORT_FILE"
echo "" >> "$REPORT_FILE"
echo "Detalle de los atacantes (Usuario e IP origen:" >> "$REPORT_FILE"
echo "-----------------------------------------------------">> "$REPORT_FILE"

grep "$TODAY_SYSLOG" "$LOG_FILE" | grep "Failed password" | awk '{print "-> Usuario atacado: " $9 " | Ip de origen: " $1
echo "-----------------------------------------------------------------" >> "$REPORT_FILE"
echo "Fin del reporrte." "$REPORT_FILE"
echo "Análisis completado!_Revisa el archivo $REPORT_FILE para ver los resultados."
```

Reto B: Configuración del "Escudo" Perimetral con UFW

1. **Reglas de Sigilo**

```
# ok icmp codes for INPUT
-A ufw-before-input -p icmp --icmp-type destination-unreachable -j ACCEP
-A ufw-before-input -p icmp --icmp-type time-exceeded -j ACCEPT
-A ufw-before-input -p icmp --icmp-type parameter-problem -j ACCEPT
-A ufw-before-input -p icmp --icmp-type echo-request -j DROP
```

2. **Reglas de Servicio**.

```
pablo@pablo:/bash/scripts$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
pablo@pablo:/bash/scripts$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
pablo@pablo:/bash/scripts$ sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
```

3. **Activación y Evidencias**

```
pablo@pablo:/bash/scripts$ sudo ufw enable
Firewall is active and enabled on system startup
pablo@pablo:/bash/scripts$ sudo ufw reload
Firewall reloaded
pablo@pablo:/bash/scripts$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)

pablo@pablo:/bash/scripts$ sudo ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     ALLOW IN    Anywhere
[ 2] 22/tcp (v6)                ALLOW IN    Anywhere (v6)

pablo@pablo:/bash/scripts$
```

4. **Prueba cruzada**